Aniketh Parlapalli

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EDUCATION

University of Washington, Electrical/Computer Engineering **Seattle, WA**

Bachelor of Science in Electrical/Computer Engineering September 2024 - Present

Sophomore in ECE

Relevant Coursework: Circuits & Electronics, Computer Hardware Skills, Signals & Systems, Data Structures

Work Experience

Kairos Technologies **Remote**

AI Intern June 2025-September 2025

- Developed Python AI agent pipelines predicting crypto markets with 80% accuracy.
- Co-authored research paper, comparing effectiveness of 6 LLMs in predicting crypto market emphasis on Sequence to Sequence models.

Husky Satellite Lab **Seattle, WA**

Co-Lead *December 2024- Present*

- Design, research, test magnetorquers to stabilize the satellite HuskySat-2, which will launch in 2026.
- Developed embedded flight software for satellite attitude control using F Prime), focusing on real-time control loops, finite state machine design, and efficient inter-component communication.
- Led the end-to-end development of embedded electronics and flight-grade PCBs, including schematic capture, microcontroller integration, mixed-signal design, and EMI mitigation, enabling precise magnetorquer control synchronized real-time firmware.

UW ROV **Seattle, WA**

Electrical Member December 2024- Present

- Designed and validated custom PCBs using ECAD tools, integrating sensor interfaces and motor control circuitry, reducing wiring complexity by 30% and improving system reliability
- Developed control software and interfaces using machine learning and computer vision, acheiving 90% object detection accuracy during simulations, and reducing risk of failure in underwater operations by 50%.
- Performed hardware-in-the-loop testing and low-level firmware debugging, validating sensor acquisition, actuator control, and communication under underwater and pressure-stress conditions

Projects

STEVE Research Project- UW Personal Robotics Lab **Seattle, WA**

November 2025- Present

- Trained a diffusion-based robotic control policy on Linux using 100+ teleportation trajectories achieving >90% task success on valve and handling manipulation, built an end-to-end AI pipeline on Linux with ML.
- Designed a real-time embedded robotics stack, integrating MCU firmware, high-rate sensor buses, and Linux edge compute streaming 100+ Hz telemetry into learning-based control loops and cutting policy convergence time by ~35%.

Line-Following Robot **Seattle, WA**

September 2025 -January 2026

- Developed embedded firmware in C/C++ on Arduino Mega, implementing a real-time closed-loop PID controller achieving <100 ms control latency and autonomous navigation.
- Interfaced with motor driver ICs via hardware PWM timers, generating duty-cycle controlled signals to drive DC motors and achieve fine-grained actuator control in a real-time system.

Autonomous Robot **Spokane, WA**

September 2024 -June 2025

- Implemented Python image-processing script on Raspberry Pi to differentiate fish from plastic debris across 200+ tests (~85% accuracy), integrating with Arduino for low-latency actuator control systems
- Designed Arduino-controlled system using 3 servo motors to actuate syringe-based ballast chambers and an ultrasonic sensor (~4 m range) to assist autonomous plastic and debris collection.

ADDITIONAL INFORMATION

Skills: Java, Python, C++, PCB Design(ECAD), Embedded Systems, Excel, Circuit Design